

Integrated electro-optics on thin-film lithium niobate

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Abstract

Electro-optics bridges electronics and photonics and serves as a foundation for a wide array of applications from communications and computing to sensing and quantum information. Integrated electro-optic approaches, in particular, enable essential electronic high-speed control for photonics while offering photonic parallelism for electronics. Recent developments in thin-film lithium niobate photonics have advanced its use for electro-optics. This technology offers not only the necessary strong electro-optic coupling but also ultralow optical loss and high microwave bandwidth. Its tight field confinement and compatibility with established nanofabrication techniques allow for excellent reconfigurability and scalability, aiding the creation of devices and systems that were deemed nearly impossible in bulk systems. Building on this platform, various new electro-optic devices^{1–16} have emerged, which surpass the current state of the art^{1–9,12–16} and introduce functionalities that previously did not exist^{3,10,11}. Thin-film lithium niobate provides a unique platform to explore various areas of physics, including photonic non-Hermitian synthetic dimensions^{17–19}, active topological physics^{20,21} and quantum electro-optics^{15,22–24}. In this Review, we present the fundamental principles of electro-optics, drawing connections between fundamental science and state-of-the-art technology. We discuss the accomplishments and prospects of integrated electro-optics enabled by the thin-film lithium niobate platform.

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Key points

- As an integrated electro-optic platform, thin-film lithium niobate enables fully integrated electronic–photonic circuits, combining the high parallelism and low latency of photonics with the high-speed control and strong nonlinearity of electronics.
- The key advantages of the thin-film lithium niobate platform include strong electro-optic interaction, low optical loss, high microwave bandwidth and scalable reconfigurability.
- Electro-optic modulation mediates interactions between different modes by adding or removing a microwave photon to or from an optical photon.
- Electro-optic interactions are governed by energy conservation, phase matching, and selection rules, with the interaction formalism varying across different modulation regimes.
- Thin-film lithium niobate uniquely integrates $\chi^{(2)}$, $\chi^{(3)}$ and electro-optic effects, which provides opportunities to explore complex nonlinear dynamics.

Introduction

Over the past century, photonic and electronic systems, both governed by the laws of electromagnetism, have become indispensable for advancing science and technology. The evolution of modern electronic systems, culminating in the creation of integrated circuits, has revolutionized our daily lives, influencing everything from computers and cell phones to automated control systems and robots. Integrated photonic systems, which offer broad bandwidth, minimal propagation loss and massive parallelism, enabled by breakthroughs in microfabrication and nanofabrication technologies, have reshaped fields such as sensing, imaging, metrology, spectroscopy and biomedicine.

At the intersection of electronics and photonics is the field of electro-optics, which harnesses distinct yet complementary properties of photons and electrons. On one end, electro-optics endows passive photonic systems with fast control provided by high-speed electronics, thus enabling modern communications^{25–29}, lidar/beam steering^{30,31}, and optical coherence tomography^{32,33} systems. Conversely, photonics enriches electronics with massively parallel processing capabilities, which has, for example, driven advances in photonic computing^{34–38} and microwave photonics^{39–43}. These application-driven developments bridge the electronic and optical domains, leveraging materials that offer control of photons using electrical signals. Although several physical mechanisms can be used, including thermo-optics, acousto-optics, free-carrier dispersion, electro-absorption and piezomechanics, the Pockels or electro-optic (EO) effect, present in materials with $\chi^{(2)}$ nonlinearity, is a clear front runner.

An ideal material platform for electro-optics needs a strong interaction between optical and electrical fields, along with minimal optical and electrical losses. It should also be amenable to microfabrication to construct advanced and complex systems in a scalable fashion. Although the search for the material that meets these stringent criteria is ongoing, an emerging photonic platform may offer a compelling solution – thin-film lithium niobate (TFLN). Having already played a pivotal role in optical technology for over seven decades, lithium

niobate (LN) is the cornerstone of various technical applications and continues to drive scientific research. Its large $\chi^{(2)}$ nonlinearity (around 30 pm V^{−1}), high refractive index (about 2.2) and wide transparency window (0.4–5.5 μm) have made it the workhorse of optical phase and amplitude modulation in modern communication technology. The fact that it can be periodically poled also makes it interesting for frequency conversion in nonlinear and quantum optics.

Despite these favourable properties of LN, the absence of high-quality thin films and difficulties associated with patterning have historically hindered its integration into complex chip-scale systems. The recent commercialization of wafer-scale thin-film LN-on-insulator^{44–46} and the development of high-quality argon-based physical etching^{47–50} have opened new possibilities for integrated electro-optics. These advances have enabled nanoscale optical confinement that result in enhanced EO interaction, while maintaining low optical losses. The history of the burgeoning field of TFLN photonics has been detailed in several comprehensive Review articles^{51–53}. Comparisons of TFLN to other material platforms have been reviewed elsewhere⁵⁴.

The essential component of TFLN electro-optics is the high-performance EO modulator. It allows electronic control of all degrees of freedom of light, including amplitude^{4,5}, phase⁵⁵, frequency³ and polarization state⁵⁶, as well as its temporal⁵⁷ and spatial profile⁵⁸. TFLN-based modulators feature a very low half-wave voltage V_π (voltage required for a π phase shift of light, typically around 1 V), low optical propagation loss ($\alpha \approx 1$ dB m^{−1}) and high EO bandwidth (of the order of 100 GHz)^{1,5,48,59}. These attributes are the consequence of a strong EO interaction, long photon lifetime and low microwave losses. In comparison to other photonic platforms, such as silicon and indium phosphide, which may excel in one or two metrics, the simultaneous achievement of all three attributes represents a substantial advantage of TFLN photonics. From a physics perspective, TFLN enables the strongest coupling between optical and microwave modes, leading to richer dynamics with enhanced interaction strength. This distinct advantage sets it apart from other EO platforms. From an application perspective, the combination of low V_π and high EO bandwidth results in energy-efficient and high-speed control. Meanwhile, minimal optical loss enables the integration of many components and/or functionalities in series within the same circuit, allowing far more operations to be performed on light than other platforms. TFLN therefore offers desirable system performance and capabilities.

Another important component of TFLN photonics is the EO resonator, which can trap photons for periods of time (photon lifetime around 10 ns for cavities with quality factors $Q \approx 10^7$)⁴⁸. This duration surpasses the operation time required to extract the photon from (or inject the photon into) a cavity using the EO effect, which is about 0.1 ns (see Box 1), thereby allowing many EO operations within one photon lifetime. Even more scientifically interesting and technologically pertinent functionalities emerge when two or more cavities are coupled together^{3,7,10,11,17,22–24}. Such configurations enable the interaction of multiple reconfigurable photonic energy levels through strong EO coupling in TFLN, a feat nearly unattainable in their bulk LN counterparts. These advances promise a new generation of complex photonic devices and chip-scale systems (Box 1) that extend beyond the capabilities of traditional EO devices, which typically rely on EO modulation within a single waveguide and/or cavity. Combining the EO effect in TFLN with the platform's inherent optical $\chi^{(2)}$ and/or $\chi^{(3)}$ nonlinearity, and with piezoelectricity, offers many additional opportunities. These features of TFLN underscore the need for more advanced mode-coupling formalisms in electro-optics, beyond the conventional formalism based on the EO tensor and index change (Box 1).

Box 1 | Formalism of electro-optic modulation

The emergence of thin-film lithium niobate (TFLN) has expanded the landscape of electro-optic (EO) devices. These devices support many frequency modes that can be coupled via the EO effect. This process of EO coupling can be understood as removing and/or adding photons from or to a particular energy level (mode). Typically, the coupling strength Ω induced by the EO effect is of the order of 1–10 GHz (refs. 3,7), much larger than the cavity decay rate κ (10–100 MHz) for resonant modes with $Q \approx 10^6$ – 10^7 . This unique combination of Ω and κ enables numerous physical phenomena and necessitates the development of formalisms to accurately describe and analyse these increasingly intricate systems.

Conventional approach

The traditional nonlinear optics formalism to treat the EO process in lithium niobate^{51,125} is based on the concept that an applied voltage introduces a phase shift to the optical field $\Delta\phi = \beta = \pi \frac{V}{V_\pi}$, where V_π represents the voltage required for a π phase shift. The term β is the modulation index or modulation depth and is used to quantify the magnitude of the phase shift, indicating how many π phase shifts are induced by the applied voltage. Consequently, light transmitted through a waveguide phase modulator, subject to a modulation signal $V_{\text{mod}} = V \cos(\omega_m t + \phi)$, acquires a phase term $E_{\text{out}} = E_{\text{in}} e^{i\beta \cos(\omega_m t + \phi)}$, which can be expanded by Bessel function: $E_{\text{out}} = E_{\text{in}} \sum_n J_n(\beta) e^{in(\phi + \frac{\pi}{2})} e^{in\omega_m t}$. Although conceptually simple, this formalism becomes complicated when dealing with systems involving multiple spatial and frequency modes coupled together.

Mode-coupling approach

Instead of using index change to represent the phase shift, the mode-coupling approach considers modulation as a coupling

mechanism between frequency modes and involves an analysis of the Hamiltonian of the system (see the figure). Here we discuss the mode-coupling approach for modulation in a single cavity (see Supplementary Information for details and other systems).

For modulation within a single cavity, it is essential to recognize three distinct regimes that require separate considerations (ω_m , modulation frequency; κ , linewidth of the mode):

- Sideband-unresolved regime: $\omega_m < \kappa$;
- Sideband-resolved regime: $\omega_m > \kappa$ but $\omega_m \ll \text{FSR}$;
- Roundtrip-resolved regime: $\omega_m \approx \text{FSR}$ or $\omega_m > \text{FSR}$;

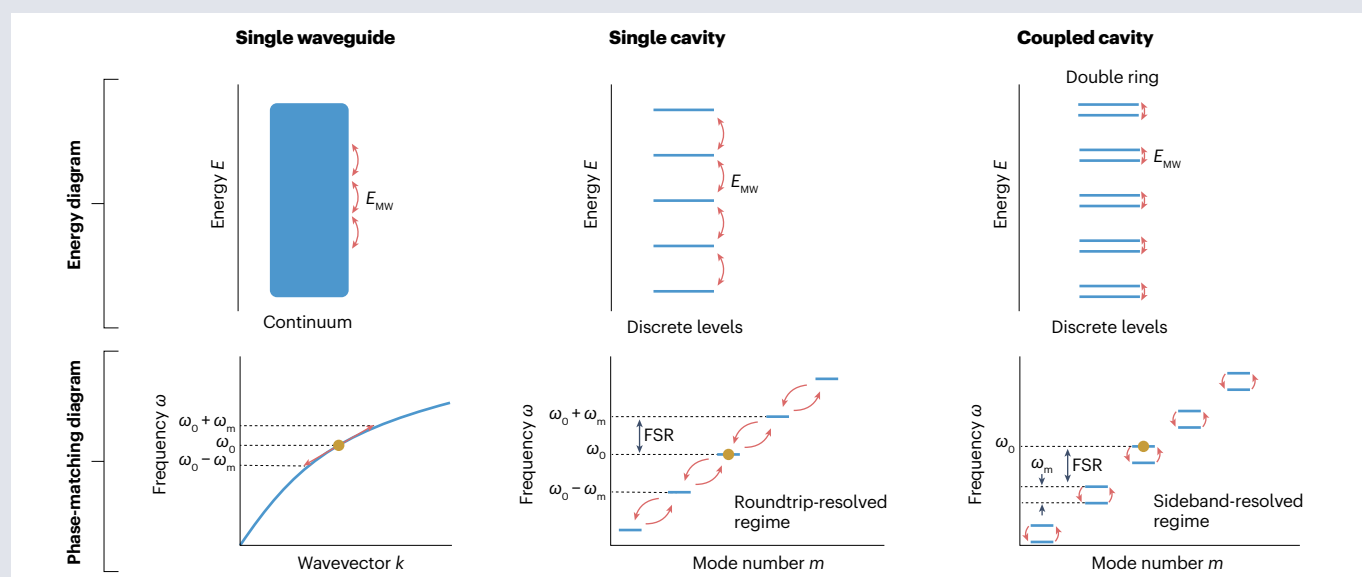
For TFLN, the ultrahigh-quality-factor resonances typically have megahertz-level linewidths, much smaller than typical modulation frequencies in the gigahertz range. Consequently, we focus on the sideband-resolved regime and roundtrip-resolved regime.

In the sideband-resolved regime, a cavity incorporating a phase modulator undergoes frequency modulation, expressed as $\omega_0 + \Omega \cos \omega_m t$. This configuration yields the Hamiltonian:

$$H = \omega_0 a_0^\dagger a_0 + (\Omega \cos \omega_m t) a_0^\dagger a_0$$

in which $\Omega = G_V \times V$ represents the EO coupling strength introduced by the microwave, a_0 is the cavity eigenmode with frequency ω_0 , and V is the peak voltage of the microwave drive. The factor G_V (approximately 0.5 GHz V^{-1} in TFLN) is related to the V_π of the modulator.

In the roundtrip-resolved regime, where the modulation period ($1/\omega_m$) is comparable to the roundtrip time ($1/\text{FSR}$), the cavity's behaviour cannot be simplified to frequency modulation because the concept of cavity frequency is established through the roundtrip interference effect. In this scenario, the EO coupling directly affects a



Box Fig. 1 | Energy diagram and phase-matching diagram for single-waveguide, single-cavity and coupled-cavity systems with microwave modulation under different modulation regimes. E_{MW} , energy of microwave photon; FSR, free spectral range, ω_m , frequency of microwave. Green dot represents an actual optical photon.

(continued from previous page)

set of optical modes $\{a_n\}$ ($n = -N, \dots, N$ with $2N+1$ being the total number of modes considered) separated by the FSR (see the figure). The Hamiltonian is given by:

$$H = \sum_{n=-N}^N \omega_n a_n^\dagger a_n + (\Omega \cos \omega_m t) (a_n a_{n+1}^\dagger + a_n^\dagger a_{n+1})$$

This Review aspires to present an introduction to TFLN electro-optics, encompassing principles, formalisms, state-of-the-art technical developments and applications. We hope that it will complement other reviews^{51–53,60–63} that provide a broad overview of TFLN devices^{51–53} or focus on nonlinear optics⁶⁰ and traditional EO modulators^{61–63}. In addition, a timeline of the development of various TFLN platforms, including related references, can be found in the Supplementary Information. We hope our work will provide insights into EO devices and systems on TFLN and foster a deeper understanding in this rapidly advancing field.

Basic principles of electro-optic modulation

Pockels electro-optic effect

The most basic physical picture for the EO effect can be visualized as a static electric field inducing a constant change Δn in the refractive index n of a material (Fig. 1a). This index change is conventionally described using the EO tensor R :

$$\begin{bmatrix} \Delta\left(\frac{1}{n^2}\right)_1 \\ \Delta\left(\frac{1}{n^2}\right)_2 \\ \Delta\left(\frac{1}{n^2}\right)_3 \\ \Delta\left(\frac{1}{n^2}\right)_4 \\ \Delta\left(\frac{1}{n^2}\right)_5 \\ \Delta\left(\frac{1}{n^2}\right)_6 \end{bmatrix} = R \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}, \quad (1)$$

in which the $\Delta\left(\frac{1}{n^2}\right)_k$ with $k = 1, \dots, 6$ describes the index change via the deformation of the index ellipsoid in conventional nonlinear optics (see ref. 64 for details). Consequently, light propagating through an EO material not only experiences a phase shift stemming from the refractive index but also undergoes an additional phase shift $\Delta\phi$, which is proportional to the index change Δn and the propagation distance L .

Electro-optic modulation

Applying a temporally varying electric field, for example a sinusoidal microwave signal, to an EO material causes the induced phase shift $\Delta\phi$ to oscillate with the amplitude of the microwave signal. This results in the generation of new frequency components higher or lower than that of the incident light, with the difference given by the frequency of the microwave signal^{65,66}. This process, known as EO modulation, is a fundamental mechanism in electro-optics. Physically, EO modulation

In the conventional approach, the coupling coefficient k between waveguide and cavity, the roundtrip loss α , and the modulation index β are all unitless, whereas κ_e , κ_i and Ω in the Hamiltonian are in hertz. The relationship between these is $k = 2\pi \frac{\kappa_e}{\text{FSR}}$, $\alpha = 2\pi \frac{\kappa_i}{\text{FSR}}$, $\beta = 2\pi \frac{\Omega}{\text{FSR}}$ (ref. 7).

involves the interaction of the microwave field with the optical field, in which the frequency of optical photons is changed by adding or removing microwave photons. A detailed formalism of EO modulation, also in terms of coupling between different energy levels, can be found in Box 1.

In terms of applications, EO modulation can bridge the microwave and optical domains, which is crucial for the global optical communication network. It can imprint analog microwave signals on an optical carrier, or it can encode digital information in both amplitude and phase of the optical signal. Conversely, optical signals with differing frequencies can interact via the EO effect, generating microwave signals that can be used to excite or control an electronic system. Conventionally, EO modulation is used to rapidly modify degrees of freedom of light, such as its temporal profile (for example by switching light on and off) or spectral content (for example by generating sidebands).

Materials and device structures for TFLN photonics

In TFLN photonics, the basic device structures are the optical waveguides used to confine the light, and the electrodes to support the microwaves. The LN film on which the waveguides and electrodes sit are created using ‘smart cut’ technology (Fig. 1b) by slicing the bulk LN along a specific crystal axis. LN is a birefringent material with two different refractive indices along different axes⁶⁷ – typically, an ordinary refractive index n_o along the x and y axes, and an extraordinary index n_e along the z axis. For LN, n_e is smaller than n_o , making it a negative uniaxial crystal. The most commonly used LN films can be classified into two types: x -cut and z -cut. An x -cut thin film is precisely cut normal to the crystal x -axis, thus extending the film in the y – z plane of the LN crystal, whereas a z -cut film extends in the x – y plane of the crystal.

An ideal device design therefore has to take the material properties into account. To make use of the largest EO tensor element, which in LN is r_{33} (approximately 31 pm V^{-1}), the electric field must be applied along the crystal z axis. The electrode used to support the microwave field is therefore designed to ensure the electrical field penetrates the slab along the z axis of TFLN. For example, in x -cut TFLN, where the waveguide is ideally defined along the y axis, a pair of metal electrodes running parallel to it forms a capacitor, or for high-speed application a microwave transmission line (see Supplementary Information), ensuring an electric field along the z axis (Fig. 1c). This configuration may not be useful or possible in all designs, such as complex resonant structures, and then only the field projection onto the z axis contributes to effective modulation. In z -cut TFLN, the transmission line is designed so that the signal is on top of the waveguide, leading to an electric field perpendicular to the film (Fig. 1c).

Another viable option is the hybrid form of TFLN, which involves the creation of a waveguide through heterogeneous rib-loading^{68–78} (Fig. 1c) or film-bonding^{4,73,79–82} (Fig. 1c). The hybrid TFLN platforms

avoid etching LN, which alleviates concerns of possible Li^+ contamination. This has led to extensive research on integrating TFLN with mature silicon and silicon nitride integrated photonics^{4,68–80,83}. Chiplet or coupon-level bonding^{4,73,79,80} and wafer-scale bonding of TFLN on Si or on SiN have been demonstrated^{81,82}. Recently, high-quality monocrystalline Si-LN-on insulator wafers have also been achieved^{70,71}. These hybrid platforms typically have weaker mode confinement in the LN layer and therefore somewhat degraded EO efficiency. Nonetheless, with optimized mode engineering, state-of-the-art hybrid EO devices are exhibiting comparable performance to the monolithic ones and may already satisfy most practical applications^{4,73,84–86}.

In this Review we focus on monolithic (etched) *x*-cut TFLN waveguides, but our discussions and conclusions qualitatively also apply to other system configurations.

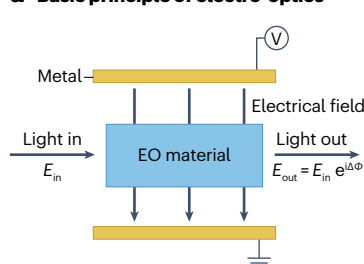
Major classes of electro-optical devices on TFLN

The three main classes of TFLN EO devices are those based on waveguides, cavities (resonators) and coupled cavities (Fig. 1d). The most

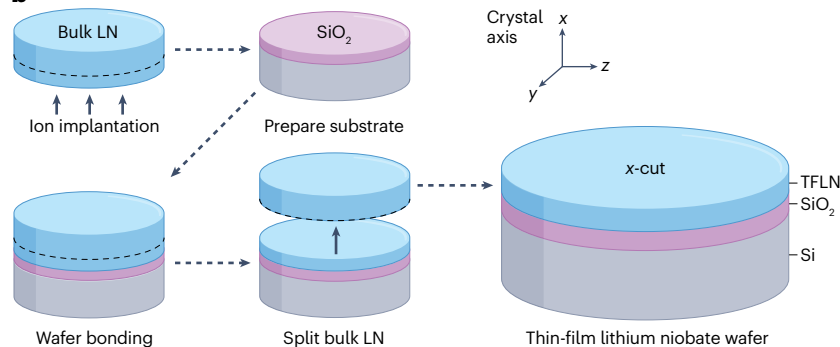
fundamental structure is the waveguide-based phase modulator⁵⁵, which allows for precise phase control of the light passing through it. By placing one phase modulator in each arm of an on-chip Mach-Zehnder interferometer, one can realize an amplitude modulator. This device leverages the interference between two paths that light can take to control the amplitude of light^{1,4,5,59}. Single-cavity devices can be achieved by placing a phase modulator inside a cavity^{2,18}. In this configuration, the phase change effectively alters the free spectral range (FSR) of the cavity, changing the frequency of the resonance. Because the EO effect typically induces an FSR change that is much smaller than the FSR itself, an external phase modulation in most cases only leads to a noticeable modulation of the resonance frequency. Alternatively, one can embed an amplitude modulator in a cavity, allowing control over the resonator's loss. It is worth noting that phase-modulated and amplitude-modulated cavities manipulate the real and imaginary components of the energy of the resonator mode.

Coupled-cavity devices are more advanced devices enabled by TFLN^{13,7,11,17}. They typically consist of two or more evanescently coupled

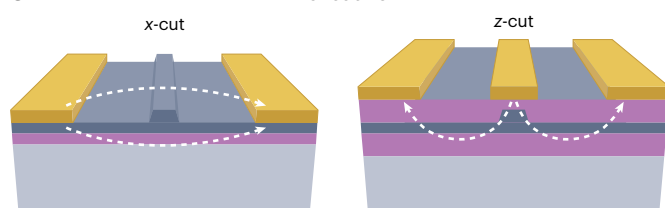
a Basic principle of electro-optics



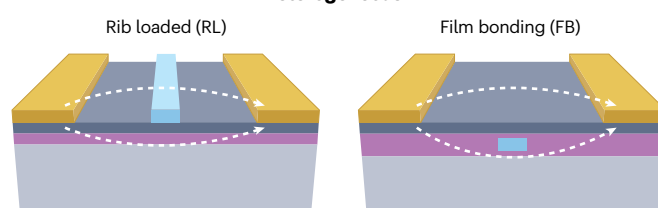
b



c Monolithic



Heterogeneous



■ Lithium niobate ■ Silicon dioxide ■ Silicon ■ Gold ■ Silicon nitride or silicon

d Basic principle of electro-optics

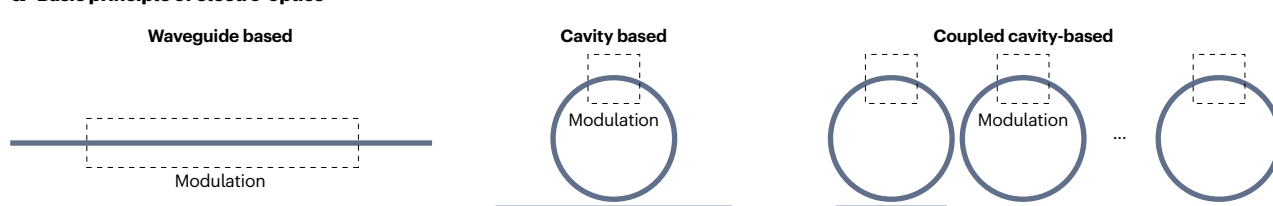


Fig. 1 | Basic principle and device structures on thin-film lithium niobate. **a**, Light sent into an electro-optic (EO) material will acquire a phase shift that is proportional to the voltage applied to the material. **b**, Illustration of a thin-film lithium niobate (TFLN) wafer and the smart cut technology used to prepare it. The coordinate axis represents the crystal axis of the lithium niobate (LN) crystal.

c, Cross-section for major types of TFLN EO platforms. The white dashed arrows denote the microwave electric field lines. **d**, Illustration of three major classes of EO devices: waveguide, cavity and coupled cavity. The dashed box is used to label the electrode region in which EO modulation can be applied.

cavities (Fig. 1d). Each cavity supports discrete optical modes (energy levels) each with its own loss or gain (such as from ion doping or parametric gain). The advancement of coupled-cavity devices on TFLN lies in the EO modulation, which enables photonic EO transitions between different modes. These levels and their associated loss or gain are highly reconfigurable. Consequently, various coupled-cavity devices with new functionalities and high performance have been realized over the past years, as discussed in more detail later.

Electro-optics enabled by thin-film lithium niobate Modulation in non-resonant waveguide devices

Waveguide-based EO devices, such as phase modulators, amplitude modulators, and in-phase and quadrature (IQ) modulators, are the most commonly used devices on TFLN (Fig. 2). They enable a wide range of

functionalities such as phase tuning, switching, comb generation and spectral control. In this section, we will discuss their basic structures, working principles and state-of-the-art developments.

Waveguide-based phase, amplitude and IQ modulators. EO phase shifters and modulators can be realized by passing a waveguide through a pair of electrodes^{55,87,88} (see Fig. 2Aa). Embedding phase modulators in an interferometer or resonator can implement an amplitude modulator. The most widely used form of amplitude modulator is the travelling-wave Mach–Zehnder modulator (Fig. 2Ba), where the two arms of a Mach–Zehnder interferometer pass through the two gaps of a coplanar waveguide electrode¹. In a Mach–Zehnder modulator, the half-wave voltage V_π is half of that of a phase modulator with the same length, as the two arms experience opposite phase shifts (Fig. 2Bb).

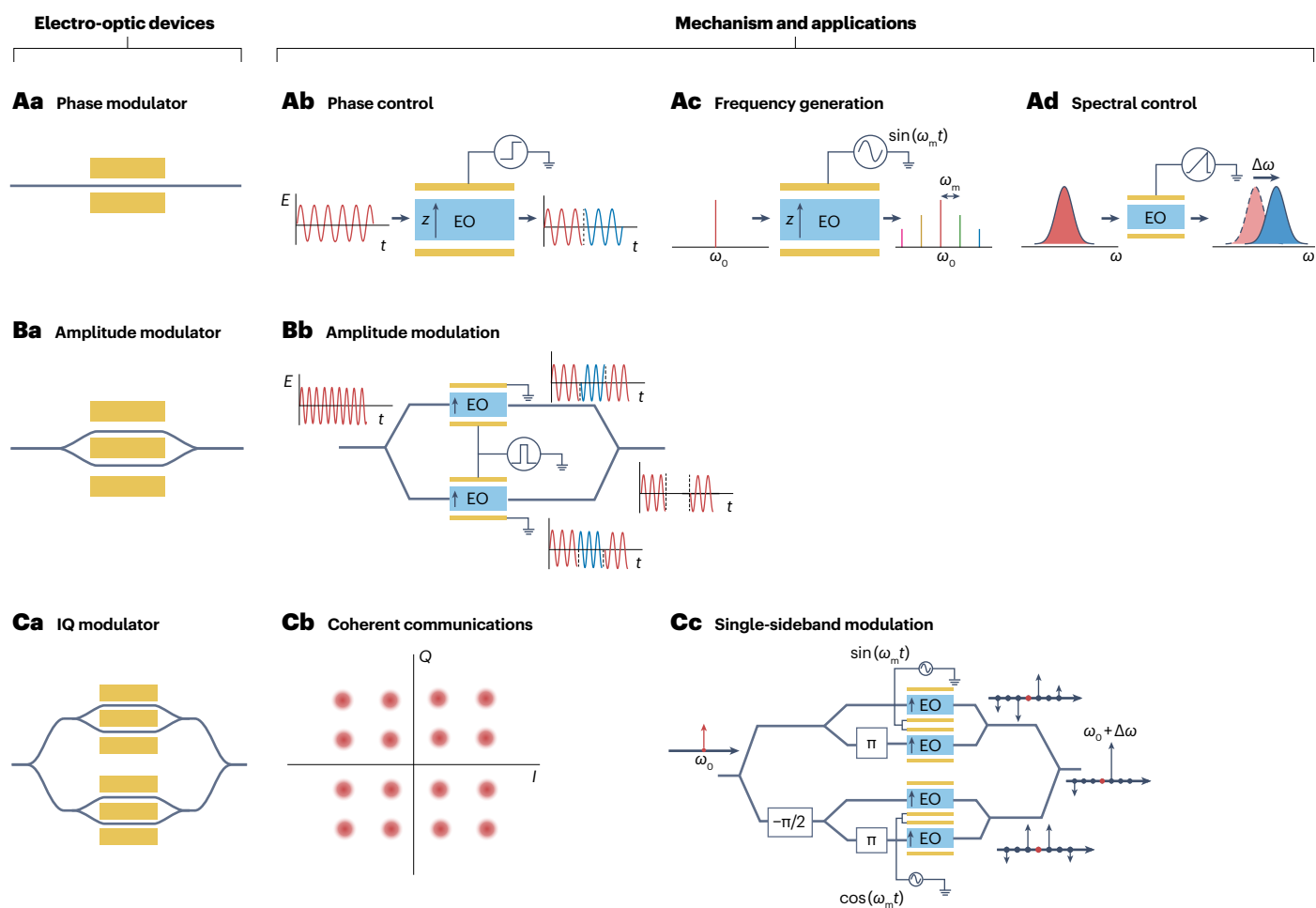


Fig. 2 | Waveguide-based electro-optic devices on thin-film lithium niobate. **Aa**, Phase modulators. Embedding a thin-film lithium niobate (TFLN) waveguide between a pair of electrodes forms a basic phase modulator. **Ab**, Applying a voltage on the electrode alters the refractive index of lithium niobate (LN) and imparts a phase shift on the output optical field. **Ac**, Driving a phase modulator with a sinusoidal radio wave creates sidebands. This configuration can also be used as a frequency mixer. **Ad**, It is also possible to achieve spectral control of input light through various dynamic modulation schemes. For example, a linearly ramping phase modulation results in a frequency shift of the input optical field. **Ba**, Amplitude modulators. Combining two phase modulators in the form of

a Mach–Zehnder interferometer creates an amplitude modulator. **Bb**, Typical TFLN amplitude modulators adopt the push–pull configuration, which creates opposite phase shifts in the two arms, to reduce the half-wave voltage. **Ca**, In-phase and quadrature (IQ) modulator. IQ modulators consist of a pair of amplitude modulators and allow independent modulation of both quadratures of the input field, which is a key component for coherent communications (panel Cb). **Cc**, When applying appropriate phase shifts or d.c. bias, the IQ modulator can achieve single-sideband modulation by cancelling unwanted sidebands, which is important for applications such as radio-over-fibre, sensing and microwave photonics.

Key figures of merit of an EO modulator include insertion loss, half-wave voltage, EO bandwidth, linearity and extinction ratio (only for amplitude modulators). These figures of merit are interrelated, and tradeoffs exist. For example, a smaller electrode gap reduces V_π but may increase metal-induced optical absorption; a longer modulator has smaller V_π , but leads to smaller EO bandwidth owing to microwave loss and possibly imperfect velocity matching. Overall, the EO performance largely depends on electrode design, in which velocity matching, impedance matching and microwave loss are important considerations (see Supplementary Information). A key advancement in improving the EO bandwidth is the adoption of capacitively loaded coplanar waveguide electrodes^{59,89–91}, in which microstructured segments are added in the gaps of the coplanar waveguide. Doing so prevents current crowding on the edge and reduces microwave loss. One drawback is that the capacitive loading reduces the microwave phase velocity, which requires a low-index handle wafer (such as quartz⁵⁹) or undercuts in the high-index silicon substrate⁸⁹ to maintain velocity matching.

State-of-the-art TFLN Mach–Zehnder modulators have achieved V_π of around 1 V, extinction ratio greater than 20 dB, 3-dB bandwidth over 100 GHz and total insertion loss of a few dB (including a large fibre-chip coupling loss and a sub-dB on-chip insertion loss)⁵. Such high-performance modulators are enabling technologies to address the exponentially expanding data traffic driven by fast-growing computing and data centre infrastructure. In particular, as optical transceiver technology^{92–95} progresses towards 800 gigabits per second (Gbps), 1.6 terabits per second (Tbps) and eventually 3.2 Tbps, 200 or 400 Gbps per lane will be required. This pushes the analogue bandwidth requirement for EO modulators beyond 100 GHz, which is difficult to achieve with silicon photonics but readily accessible with TFLN^{1,5,73,83,96,97}.

Combining two Mach–Zehnder modulators can realize an in-IQ modulator⁹⁸ (Fig. 2Ca) – an essential component for coherent communications (Fig. 2Cb). With a dual-polarization IQ modulator, a record-high single-wavelength data rate of 1.96 Tbps was demonstrated^{5,99}. With proper phase setting in each beam path of the interferometer, an IQ modulator can achieve optical single-sideband modulation by suppressing unwanted sidebands (Fig. 2Cc). Optical single-sideband operation is important for applications such as radio-over-fibre, microwave photonics, sensing and frequency-modulated continuous wave lidar. Some more recent developments include meander designs of long modulators^{100,101}, investigation of millimetre-wave operation⁹⁶, improved linearity using a ring-assisted Mach–Zehnder interferometer configuration¹⁰², and double-pass or multipass (or loop-back) phase modulators for radiofrequency V_π reduction^{15,103}.

The transparency of LN at visible wavelengths renders TFLN modulators compatible with technologies ranging from biological and environmental sensing to augmented or virtual reality, and quantum information science and technology. Owing to the wavelength dependence of the EO phase shift $\Delta\phi \sim \lambda^{-1}$ and the reduced waveguide–electrode gaps at shorter wavelengths^{13,14}, amplitude modulators with $V_\pi L_\pi$ as low as 0.17 V cm at 450 nm have been reported¹⁴, for which the highest reported 3-dB bandwidth (around 35 GHz) was limited only by the electrode design^{13,104}. Modulation with extremely high extinction ratio (45 dB) has also been demonstrated at visible wavelength using frequency upconversion¹⁰⁵. There are also reports of quantum applications of TFLN visible modulators, including scalable processing of single photons from a deterministic quantum emitter¹⁰⁶ and photonic

engines integrated with spatial light modulators for fast, free-space control of atomic systems¹⁰⁷.

The TFLN platform supports various microstructures and nanostructures that allow more complex modulator designs. This includes resonant modulators based on micro-rings^{3,108–110} or photonic crystals^{111,112} as well as Bragg-grating-based reflector or Fabry–Pérot cavities^{113–115}. Resonant modulators have small footprint and low V_π , but high-Q resonances will limit their operating bandwidth. One potential resolution is to use a design where a ring resonator is coupled to the bus waveguide through a Mach–Zehnder modulator, and modulation affects the coupling instead of shifting the resonance¹¹⁶. Some other variations of EO modulators include Michelson-interferometer modulators^{117–119}, EO polarization controllers¹²⁰ and compact slow-light Mach–Zehnder modulators¹²¹.

TFLN modulators offer substantial improvements over previous commercial LN modulators and devices on other material platforms (a detailed performance comparison can be found elsewhere^{1,62}). Further improvements of V_π , bandwidth and insertion loss are possible by increasing the microwave–optical mode overlap or by improving the optical loss, velocity matching and the microwave loss (see Supplementary Information). Details about the limitation of these properties can be found in our previous Review⁵¹.

For large-scale EO systems, it is crucial to reduce the TFLN modulator size, which can be achieved by folding the microwave electrode or exploring new types of modulators. TFLN EO devices also suffer from d.c. drift – a long-standing challenge in this field.⁵¹ Recently it has been shown that thin-film lithium tantalate may offer better d.c. stability than TFLN^{122–124}. The widespread industrial adoption of telecom-wavelength TFLN phase and amplitude modulators has accelerated their development, and the ever-expanding literature is difficult to capture fully here. This section only offers a brief overview of some representative works and relatively recent progress. For a more thorough and detailed review on this specific topic, readers may refer to papers elsewhere^{51,62,63}.

Waveguide-based electro-optic frequency comb generation. The EO frequency comb is generated by microwave-driven modulators, resulting in sideband generation where each sideband represents one comb line (Fig. 2Ac). It offers electrical control over the comb dynamics and maintains intrinsic phase coherence among all comb lines. A comprehensive review on EO combs can be found elsewhere¹²⁵. Traditional EO combs are typically waveguide based. In this setup, light passes through a phase modulator, and the output light can be expressed as $E_{\text{out}} = E_0 e^{i\omega_0 t - ikz} e^{i\beta \cos(\omega_m t + \phi)}$. The term $e^{i\beta \cos \omega_m t}$ is the generation function of a Bessel function. Therefore, one can directly expand $e^{i\beta \cos \omega_m t}$ to obtain the analytical solution for the output field: $E_{\text{out}} = E_0 e^{i\omega_0 t - ikz} \sum_n J_n(\beta) e^{in(\phi + \frac{\pi}{2})} e^{in\omega_m t}$. Building on this, one can cascade several phase or amplitude modulators together to obtain a broader waveguide EO comb¹²⁶, in which the use of amplitude modulators flattens the comb shape.

The field of waveguide EO combs has rapidly grown in recent years. Frequency combs with more than 40 spectral lines spanning 10 nm have been demonstrated using only a single phase modulator⁵⁵ with $V_\pi = 3.5$ –4.5 V at 5–40 GHz. A microwave driving power of the order of 3.1 W is used, corresponding to a modulation index $\beta \approx 4\pi$ (ref. 55). Building upon this breakthrough, flat-top waveguide EO combs on TFLN were first demonstrated by cascading an amplitude modulator with a phase modulator^{12,127}. For example, in ref. 12, a total of 67 comb lines with a 12.6-nm span were demonstrated by driving the device with

3.8 W microwave power. This work used a loop-back structure for the phase modulator to reduce the V_π of the phase modulator to 2–2.5 V. The loop-back geometry, however, creates a weak resonance that makes the V_π frequency dependent. Although bulk systems can achieve similar comb line numbers and spans, they require a cascade of one amplitude modulator with three phase modulators. Recently, cascaded amplitude and phase modulator frequency combs using a single microwave drive have been demonstrated on TFLN, reducing the complexity of the system¹²⁸. Another drawback of these EO comb sources, as well as their bulk counterparts, is their reliance on multiple high-power microwave sources. But these waveguide-based combs are ideally suited for applications, such as data communications, that require good spectral flatness and do not need a wide span.

Waveguide-based frequency conversion with spectral shearing.

EO spectral shearing is another approach to alter the frequency of light (Fig. 2Ad). It enables a substantial shift in frequency, typically of the order of 100 GHz, and requires only a single-tone microwave driving signal. The underlying principle of spectral shearing is the fact that a linearly ramped phase modulation $\Delta\phi = Kt$ is equivalent to a frequency shift of light: $E_{\text{out}} = E_{\text{in}}e^{iKt} = E_0e^{i\omega_0 t - ikz}e^{iKt} = E_0e^{-ikz}e^{i(\omega_0 + K)t}$. Importantly, although spectral shearing can achieve a very large frequency shift beyond the microwave driving frequency (for a sinusoidal microwave drive), it only works for pulsed light. The duration of the pulse needs to be shorter than the linearly rising (or falling) part of the sine wave. When using a sinusoidal modulation, the applied phase modulation is given by $\Delta\phi = \beta\sin(\omega_m t + \Delta\phi) \approx \beta\omega_m t \cos\Delta\phi$, in which $\Delta\phi$ is the relative phase between the microwave and optical pulse, ω_m is the microwave frequency and β is the modulation index¹⁵ (Box 1). Here, the frequency shift $\Delta f = K = \beta\omega_m \cos\Delta\phi$ of the linear ramping phase represents the magnitude of the frequency shift and is determined by the peak-to-peak voltage and the frequency of the sinusoidal signal¹⁵.

An important figure of merit is the effective shift magnitude ($\Delta f/\delta f$) that is defined as a ratio between the frequency shift that can be achieved, Δf , and the bandwidth of optical pulse, δf . Consequently, increasing the frequency of the modulation signal to obtain a larger Δf (as $\Delta f \propto \beta\omega_m$) does not necessarily increase $\Delta f/\delta f$. Higher-frequency modulation leads to shorter periods, necessitating shorter pulse durations (higher pulse bandwidth δf). As a result, the ultimate limit on $\Delta f/\delta f$ is determined by the amplitude of the applied phase modulation, which is constrained by V_π of the phase modulator. Although a sawtooth modulation^{129,130}, in which the voltage is repeatedly linearly ramped, mitigates the bandwidth constraint, the necessary high-bandwidth microwave signal is difficult to achieve, even with high-bandwidth modulators.

Spectral shearing has been successfully demonstrated in bulk systems^{131,132} and used in quantum applications but typically has limited $\Delta f/\delta f < 1$ with $\Delta f \sim 200$ GHz, owing to the high V_π of bulk modulators. On TFLN, spectral shearing of single photons with a shift frequency of $\Delta f = \pm 641$ GHz (± 5.2 nm) has been achieved¹⁵, resulting in $\frac{\Delta f}{\delta f} > 1$. Furthermore, applying quadrature phase modulation to the same device has achieved an impressive 8-fold suppression of pulse bandwidth (from 6.55 nm to 0.35 nm). Using TFLN modulators with sub-volt π -voltage at visible wavelengths, a spectral shearing with $\Delta f/\delta f > 7$ has been demonstrated with $\Delta f = 6$ GHz (ref. 13).

Modulation in single-cavity devices

The TFLN platform makes it possible to combine strong EO effects with microcavity enhancements. Common cavity geometries considered in

TFLN are Fabry–Pérot and microring cavities (Fig. 3Aa,Ba). These device structures underpin several critical applications including modulation, dispersion engineering, comb generation, synthetic crystal generation. In this section, we will discuss single-cavity-based devices for different functionalities.

Single-cavity electro-optic modulator. The Fabry–Pérot EO cavity is commonly used in free-space optics, using highly reflective mirrors (Fig. 3Ab). On a TFLN chip, these mirrors can be realized using photonic crystals, Bragg gratings or Sagnac loops. By engineering the dispersion and reflection spectrum of these mirrors, one can achieve a Fabry–Pérot cavity with a desired optical bandwidth at the target wavelengths. For example, a photonic crystal mirror reflects light whose frequency falls within the bandgap of the photonic crystal while transmitting frequencies outside the bandgap (Fig. 3Ac). Instead of using reflective mirrors, a microring cavity uses interference effects within a closed-loop waveguide to enhance a resonance. This type of cavity typically has resonances separated by the FSR over a wide wavelength range, which is critical for applications that can benefit from multiple resonant modes, such as comb generation.

Synthetic electro-optic crystal generation. The concept of a photonic synthetic dimension^{133–140} in the frequency domain derives from the equidistant arrangement of distinct optical frequencies, which form a lattice along the frequency axis. The high frequency and wide bandwidth of light, along with reconfigurability, scalability and the capability to control gain and loss, make synthetic dimensions a promising approach to simulate complex physical systems and explore novel physics. For example, they have already been used to explore non-Hermitian^{135,136}, high-dimensional^{18,141} and topological systems^{142–146} as well as moiré lattices¹⁴⁷, which can be challenging to study using other methodologies.

Nonlinear processes, such as EO modulation, can be used to create synthetic dimensions. The simplest configuration of a synthetic EO crystal involves a single EO modulation applied to a cavity with microwave frequency equal to the cavity's FSR (Fig. 3B), which is formally described in the roundtrip-resolved regime (see Box 1). Advanced structures that involve multiple modulation tones or coupled cavities can be extended in this formalism. Although simple in concept and geometry, synthetic dimensions have shown great potential and excellent experimental progress in fibre optics systems¹³⁶, but this has been limited by scalability and challenges around high insertion loss inherent to this approach. TFLN may prove a catalyst advancing the field: its strong EO effect enables strong coupling rates and low loss between lattice points, orders of magnitude beyond the loss rate of each lattice point (EO coupling $\Omega \sim 10$ GHz, optical loss rate $\kappa \sim 100$ MHz) – a regime that is hard to access on other photonic platforms. These features suggest that large-scale and intricate synthetic crystals can be realized using TFLN.

Recently, high-dimensional frequency crystals extending up to four dimensions have been demonstrated using a single EO cavity on TFLN¹⁸. Instead of folding a 1D lattice to achieve higher dimensionality, as is done conventionally^{146,148,149}, the high dimensionality was achieved by using several different microwave drives with their frequencies closely matching (detuned by 1 MHz) the cavity FSR (around 10 GHz). As a result, each microwave drive provided an individual dimension for the photons with each dimension encompassing over 100 lattice points. Furthermore, a crystal with a frequency domain mirror, which is a device capable of reflecting optical energy propagation along the frequency dimension, was proposed and experimentally demonstrated¹⁷. This was

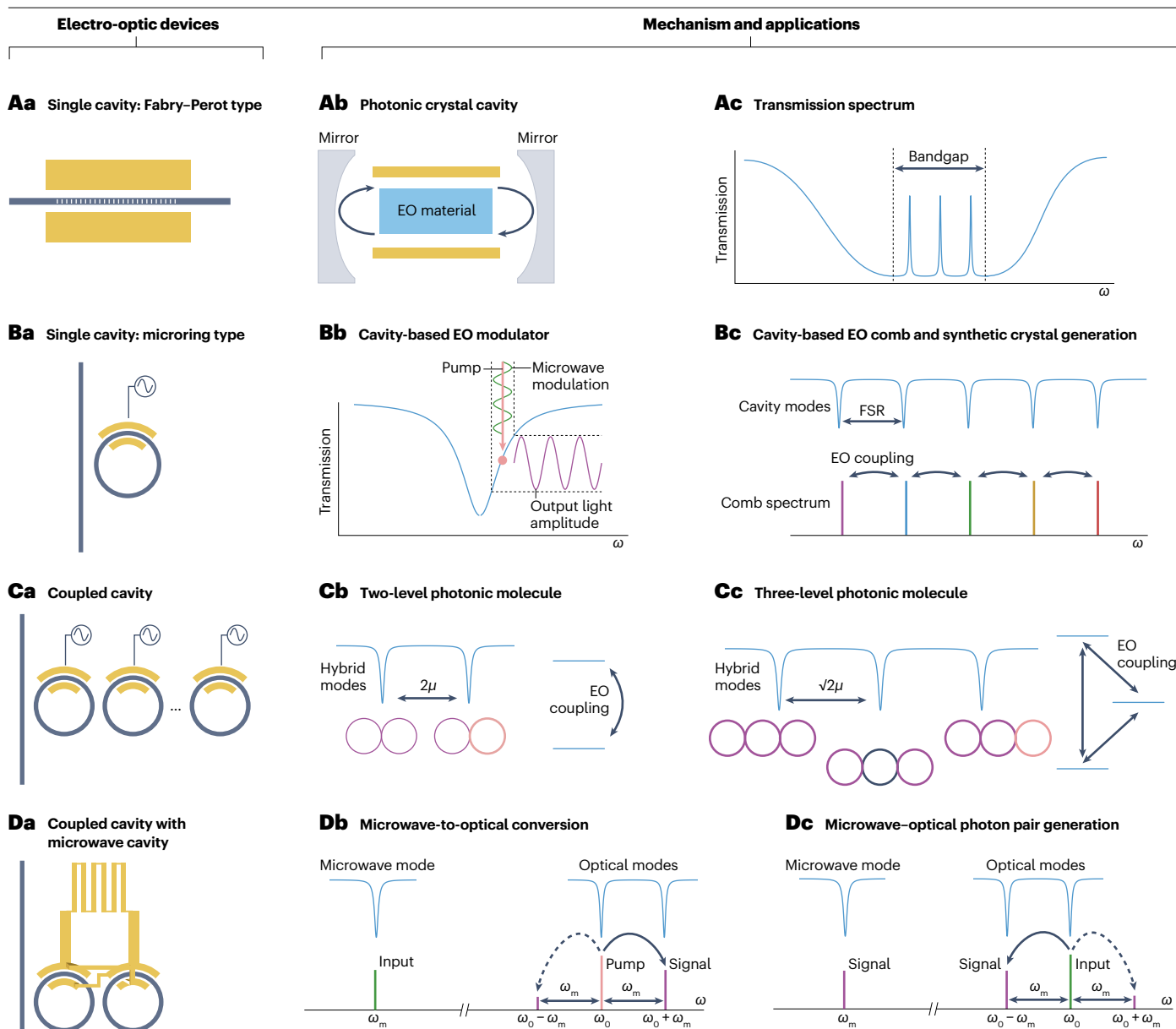


Fig. 3 | Cavity-based electro-optic devices on thin-film lithium niobate. Fabry-Pérot type single-cavity device (part A). The schematic configuration shows a Fabry-Pérot type single cavity on thin-film lithium niobate (panel Aa). The two photonic crystals act as mirrors to form a cavity with an electro-optic (EO) material inside (panel Ab). By engineering the dispersion of the photonic crystal, one can control the position of the resonances by varying the bandgap of the photonic crystal^{111,112,250–253} (panel Ac). Microring type single-cavity device (panel Ba). A cavity-based modulator (panel Bb) typically achieves amplitude modulation by varying the cavity resonant frequency. The EO cavity can also be used for frequency comb generation^{2,7,254} and synthetic crystal generation^{17–19} (panel Bc). A

coupled-cavity device (panel Ca) can be used to construct multilevel photonic molecule for frequency manipulation^{3,10,11}, such as two-level (panel Cb)^{3,11} and three-level (panel Cc) photonic molecules¹⁰. They have been used for frequency shifting and optical isolation. Coupled cavity with microwave cavity (panel Da)^{22–24,151–155,165–168,170}. The microwave cavity is resonant with the mode separation of the optical modes. In this case, both microwave and optical mode are resonantly enhanced for EO interactions, forming a cavity EO system. By pumping on the red (panel Db) and blue (panel Dc) detuned case, microwave-to-optical conversion (panel Db) or microwave-optical photon pair generation (panel Dc) can be realized^{166,167}. FSR, free spectral range; μ , coupling rate between pairs of cavities.

accomplished using both coupled-cavity and polarization crossing. The mirror had a reflectivity exceeding 0.9999, enabling interference of forward and backward waves, along with the observation of trapped states in the frequency domain. More recently, synthetic dimensions

using TFLN have extended into the quantum regime, where it was used to explore correlations between entangled photons and multilevel Rabi oscillations, in which quantum frequency conversion was used to create a second dimension¹⁹.

Cavity-based electro-optic frequency comb generation. The simplest configuration of a cavity EO comb involves placing a phase modulator inside a high- Q cavity. Light circulates within the cavity, passing through the phase modulator multiple times, which, as discussed in Box 2 (selection rule), enables broadband EO comb generation. Tight confinement of the waveguide in TFLN, along with a high-performance phase modulator, allows this configuration while maintaining a high- Q cavity. This is a considerable advantage compared with bulk systems where the insertion loss of bulk components is typically large. The underlying mechanism of single-cavity EO combs can be modelled using roundtrip-resolved modulation (see Box 1), the same model used to study EO synthetic crystals.

In 2019, a single-cavity EO comb was successfully demonstrated on TFLN, achieving a span (80 nm) one order of magnitude larger than previous EO combs^{2,150}. However, despite this impressive improvement in span, the conversion efficiency of this type of comb remains limited, approximately 0.3%⁷. The low conversion efficiency arises from the cavity becoming strongly under-coupled when driven

with a strong microwave tone. The microwave drive introduces a broadband comb generation process, acting as an effective intrinsic loss κ_{MW} for the pump mode⁷. This behaviour is analogous to the generation process for other types of frequency combs, such as Kerr combs. On TFLN, the effective loss rate owing to the microwave drive (κ_{MW}) is about 100 times higher than the external (κ_e) and intrinsic (κ_i) loss rates of the cavity ($\kappa_{\text{MW}} \approx 10$ GHz, $\kappa_e \approx \kappa_i \approx 100$ MHz for $Q \approx 1$ million⁷). This means that $\kappa_e \gg \kappa_i + \kappa_{\text{MW}}$. Simply increasing the external coupling to achieve critical coupling with the microwave drive results in high loss for resonance modes, reducing the overall bandwidth.

This issue has been addressed by applying the concept of general critical coupling, which was proposed in ref. 3 and is discussed more in the following section, to a coupled-resonator EO comb. This led to a 100-fold improvement in efficiency (30%) and a 2.2-fold improvement in span (132 nm)⁷. Another challenge with cavity EO combs is a flat-top comb shape. Recent work has shown that it can be achieved by using a boundary as a frequency-domain mirror¹⁷.

Box 2 | Fundamental rules for efficient electro-optic coupling

Although electro-optic (EO) modulators are conventionally treated as switch or pulse generators in the temporal domain or as sideband generators in the frequency domain, the EO modulation on thin-film lithium niobate (TFLN) for advanced EO functionalities (such as cascaded frequency shifting, topological transport in cavity-based devices, frequency comb generation, synthetic dimensions) can be more conveniently described as a process of EO coupling between different modes. Designing these systems using the EO mode-coupling formalism, one has to account for three fundamental rules: energy conservation, momentum conservation and selection rules.

Energy conservation

If the input light is at a frequency ω_{L1} and the goal is to generate a frequency component ω_{L2} , a microwave signal with a frequency of $\omega_m = |\omega_{L2} - \omega_{L1}|$ is required. In the case of a resonant structure, the frequencies ω_{L1} and ω_{L2} must also match the frequencies of the two individual resonant modes of the device, ω_1 and ω_2 , to make the modulation process efficient.

Momentum conservation

In all-optical three-wave mixing, the momentum conservation (also called phase matching) between three participating optical modes is an important consideration. In electro-optics, one of the optical modes is replaced with an electromagnetic signal of a lower frequency, typically in the microwave range. Given the relatively long wavelength of microwaves, their momentum ($k_{\text{MW}} = 2\pi/\lambda_{\text{MW}}$) is typically small. As a result, momentum conservation considerations are only relevant at high frequencies (above 10 GHz) in which the k_{MW} is not negligible. Additionally, momentum conservation becomes significant only when the microwave electrode has a travelling-wave configuration (see Supplementary Information for details), as standing wave lack net momentum. Consequently, this momentum conservation effect is particularly relevant for waveguide-based travelling-wave EO modulators. In modulator design, this momentum

consideration is often referred to as velocity matching (refer to Supplementary Information for details).

Selection rule

Similar to atomic physics, where an atomic transition is only allowed when an external drive induces a non-zero off-diagonal coupling between the two states in the Hamiltonian, EO transitions adhere to a selection rule: there should be a non-zero overlap between the initial and final optical modes in the presence of an EO interaction.

For example, consider the photonic two-level system formed by a double-ring coupled cavity (see Box 1), which is of interest for realizing frequency shifters and beam splitters³, quantum transducers^{22,23} and other devices. Here, EO modulation applied to the device can drive a transition between the two eigenmodes, in this case symmetric and antisymmetric modes. The phase accumulation due to the EO effect on each ring needs to be different. An identical phase advance (or delay) does not create mode overlap and only results in a global phase shift for the symmetric or antisymmetric mode (see Box 1).

Another example is the cavity EO comb, which uses an EO-modulated racetrack-style cavity in the roundtrip-resolved regime (see Box 1 for the definition of modulation regimes). This setup aims to establish coupling between modes separated by a cavity free spectral range (FSR). These different modes experience different overall phase accumulation after light completes one roundtrip in the cavity. In the roundtrip-resolved regime, which we define as $\omega_m \approx \text{FSR}$ or $\omega_m \gg \text{FSR}$, the microwaves cannot be regarded as unchanged during one roundtrip of light. Consequently, the electrode is designed such that, at a fixed point in time, the upper arm of the cavity has a phase advance (delay) while the bottom arm has a phase delay (advance). This configuration ensures that light experiences the same phase advance (delay) in both arms because the sinusoidal microwave voltage reverses its sign after half a roundtrip. Thus, based on the specific functionality of the devices, the selection rule must be tailored to create mode-overlapping coupling and needs to be carefully considered for different modulation regimes.

With the notable progress in TFLN-based cavity EO combs, we anticipate that they will become valuable broadband and stable comb sources, enhancing applications in telecommunications, metrology, astronomy, spectroscopy and more. However, cavity-based EO combs are limited by the microwave power consumption and the tunability of the repetition rate because of the resonant enhancement condition. Using a microwave cavity can significantly reduce the microwave power consumption¹⁵⁰. TFLN has provided a versatile platform for cavity-based EO combs with new structures, opening possibilities to further improve the bandwidth, conversion efficiency and spectral flatness. Finally, it is important to note that, like conventional EO combs, cavity-based EO combs also generate femtosecond pulses⁷, benefiting applications in ultrafast optics.

Modulation in coupled-cavity devices

The TFLN platform holds one unique advantage: it enables advanced structures with strong, fast EO control, beyond conventional single waveguide and single cavity. This is hard to realize in other photonic platforms, as they typically lack a strong EO effect (SiN and SiO₂), have high loss (silicon or III–V materials) or are not scalable (bulk LN). TFLN opens the possibility of combining electro-optics with devices such as coupled cavities, cavity arrays or multi-waveguide-cavity system (Fig. 3C,D). These device structures can be beneficial to a wide range of functionalities and applications, including frequency manipulation and conversion, and interconnection between superconducting qubits and optical photons. In this section, we will discuss coupled-cavity-based devices.

Coupled-cavity electro-optic modulator and its integration with a microwave cavity. Coupled-cavity systems allow nonlinear optical interaction to mediate the transition between the multiple energy levels of the system. Cavities are typically evanescently coupled – that is, photons from one cavity can jump into another cavity (Fig. 3Ca). This coupling is typically static, reciprocal and conservative. Controllable coupling can be accomplished by leveraging the strong EO effect of TFLN by integrating optical cavities with microwave waveguides or resonators. The latter is particularly interesting as a triple resonance forms a microwave–optical resonant system (Fig. 3Da). However, owing to the relatively high loss of metals used in chip-scale microwave cavities, these microwave–optical resonant systems typically rely on superconducting microwave circuitry to realize high-Q microwave cavities, at the expense of requiring operation at cryogenic temperatures^{22–24,151–155}.

General critical coupling, frequency conversion and photonic molecules. Microwave-enabled control of light holds great promise for applications requiring ultrafast operations – such as frequency conversion, sensing and switching – owing to its compatibility with ultrafast electronics and the stability of microwave signals. However, this typically necessitates a photon lifetime significantly longer than the time required to transition between states, which was difficult to achieve in other photonic platforms. TFLN overcomes this limitation by simultaneously featuring strong EO coupling of the order of gigahertz, and high-Q cavities corresponding to low loss rates of 10–100 MHz (see Box 1). Consequently, TFLN now facilitates EO transitions between modes of different energy in both single and coupled cavities. For example, within a single cavity, applying a microwave with a frequency equal to the FSR creates EO transitions across a synthetic frequency lattice. In a coupled-cavity system, microwaves can induce transitions

between several hybrid modes, if energy conservation, phase matching and selection rules are satisfied.

Recently, a photonic two-level system with EO transition was demonstrated using two coupled cavities on TFLN^{3,11} (Fig. 3Cb). This system can induce EO coupling (with $\Omega \approx 5$ GHz) between the symmetric and antisymmetric modes of the coupled cavity (see Box 1), leading to Rabi oscillations between the two energy levels. As an optical two-level system, this device is well suited for gigahertz-scale frequency control, a critical energy band for communications and computing.

Typically, this EO coupling results in energy oscillations between the two frequencies, which is unsuitable for efficient energy conversion. To address this, the concept of general critical coupling has been proposed for efficient energy conversion³ (see Supplementary Information for detailed introduction). Combining general critical coupling with the two-level system coupled to a continuum, for example a bus waveguide, enables novel EO frequency shifters and beam splitters (Fig. 3Cb). Record-breaking frequency shifting performance was demonstrated, with over 99% shift efficiency (shift ratio), 0.45-dB on-chip insertion loss and a shift frequency range of 10–30 GHz, all achieved using only a continuous single-tone microwave³.

Furthermore, the concept of cascaded frequency shifting allowed the demonstration of a cascaded frequency shifter (a triple-coupled-cavity device) that achieved a four-step cascade to a 120-GHz frequency shift using a 30-GHz microwave³. Here, energy flows through a ladder of energy levels without any back reflection. This device showcases the potential to access energy bands exceeding 100 GHz using only a low-frequency microwave source. High-performance frequency shifting and beam splitting operations in the gigahertz band have the potential to affect applications in communications^{25,156}, classical information processing^{39,43,157,158} and quantum computing^{159–164}. Finally, another type of triple-coupled-cavity system that forms a three-level system¹⁰ (Fig. 3Cc) was used to realize an optical isolator. Nearly 40 dB of optical isolation was achieved using approximately 75 MW of microwave power.

These novel EO transitions in coupled-cavity systems offer a fully reconfigurable photonic system with ultrafast control, revitalizing the applications of EO systems. Further increasing the mode overlap and the cavity quality factor can substantially increase the EO coupling regime, and extending to large-scale EO cavity arrays can enable advanced complex systems with collective dynamics.

Electro-optic interface at cryogenic temperature. The EO effect in TFLN persists at cryogenic temperatures. This offers an effective solution for realizing photonic links for superconducting microwave circuits¹⁶⁵ or for on-chip optical routing and processing for optically active ions or defects in the solid state. Superconducting devices, such as qubits, logic circuits and detectors, rely on cryogenic operation. As systems scale up, the radiofrequency cables required to address the devices will induce unmanageable heat load to the host cryostats. In addition, for constructing a quantum network that connects distant superconducting qubits, direct transmission of microwave photons through radio-frequency cables at room temperature is infeasible owing to microwave loss and thermal noise. A promising solution is to replace metallic cables with optical fibres, which have low thermal conductivity and wide bandwidth (in the terahertz range) and optical channels with ultralow loss (below 0.2 dB km^{−1}). Central to this solution is an efficient and cryogenically compatible mechanism to transduce between microwave and optical signals. Combining a TFLN EO modulator with superconducting electrodes provides a viable solution.

One approach to achieve an ultra-efficient cryogenic EO interface is to use cavity systems^{166–169}. This scheme – called cavity EO – is suitable for coherent, bidirectional microwave-optical quantum transduction^{155,170}. On TFLN, it can be implemented by replacing the electrode of a double-ring coupled-cavity modulator with a superconducting resonator (Fig. 3Da), with a resonance frequency matching the doublet splitting of the symmetric and antisymmetric optical modes^{22–24,169} (see Box 1). A strong laser pump at the low-frequency mode can upconvert a microwave photon into an optical photon at the high-frequency mode.

In cavity EO systems, the key figure of merit is the cooperativity, $C = 4g_0^2 n_p / \kappa_m \kappa_o$, which depends on the EO interaction strength g_0 , microwave loss κ_m , optical loss κ_o and the number of pump photons n_p that can be put into the system without inducing quasiparticles in the superconductor, creating noise photons or overheating the cryostat^{153,155}. State-of-the-art cavity EO transduction on TFLN has achieved an efficiency of 1.02% (internal efficiency of 15.2%)²⁴. Further improvement requires microwave and optical cavities with higher quality factors, as well as better optical coupling and cryogenic packaging.

Even with low efficiency, cavity EO transducers can enable heralded entanglement between distant superconducting systems¹⁵⁴. Pumping the cavity EO device at the high-frequency hybrid mode can generate pairs of microwave and optical photons via spontaneous parametric downconversion. The optical photons can be sent to a distant location to establish entanglement with another pair source through the DLCZ protocol. Similarly, difference frequency generation between two optical fields can be used to generate microwave signals inside the cryogenic environment, which can be used to control superconducting qubits¹⁵¹.

Cavity systems enhance EO interaction at the expense of operating bandwidth. Applications such as rapid single-flux-quantum circuits, superconducting nanowire single-photon detectors and frequency-multiplexed superconducting qubits call for a broadband, low- V_π modulator. This can be achieved using a long travelling-wave modulator with superconducting electrode, which has negligible ohmic loss and breaks the tradeoff between length and EO bandwidth. Recently, a superconducting (niobium) TFLN Mach–Zehnder modulator with effective modulation length of 1 m and a V_π as low as 42 mV was demonstrated. Such a device (with 20-cm effective length, and with 230 mV V_π and 17 GHz bandwidth) is used to transmit rapid single-flux-quantum signals (5 mV peak-to-peak voltage) through telecom optical fibres¹⁵² from 4 K to room temperature.

Heterogeneous integration with thin-film lithium niobate

All materials have their advantages and limitations. Heterogeneous integration aims to combine different materials and components in a single system to deliver full functionalities beyond what is possible with an individual material.

One main motivation for TFLN heterogeneous integration is the difficult etching process of LN and its complementary metal–oxide–semiconductor (CMOS) incompatibility. Contamination by Li^+ is a concern for CMOS foundries, and dedicated tools or chambers may be needed for adopting LN in existing fabrication facilities. Furthermore, wafer sizes and costs could become limiting factors if demand increases. Currently, 6-inch TFLN wafers are readily accessible and are being adopted by several startup companies in the field. However, modern foundries mainly have 8-inch or 12-inch fabrication lines.

Although 8-inch TFLN wafers have been developed, 12-inch wafers are not yet available because bulk LN wafer supply is lacking and market demand is low. These limitations are technological hurdles rather than fundamental limits, which can be resolved once the industry identifies a proper market and demand rises. Similarly, cost per wafer is largely influenced by production volume and will reduce in mass production.

To overcome the current limitations and challenges in large-scale TFLN fabrication, extensive research has been conducted on integrating TFLN with mature silicon and silicon nitride integrated photonics^{4,68–80}. On the fabrication side, both coupon-level^{4,73,79,80} and wafer-scale bonding of TFLN on silicon or SiN have been demonstrated^{81,82}. Recently, high-quality monocrystalline Si–LN-on-insulator wafers have also been developed^{70,71}. On the device performance side, compared with directly etched TFLN (monolithic) modulators, TFLN–Si/SiN (hybrid) modulators usually have weaker mode confinement in the LN layer and therefore lower EO efficiency. Nonetheless, with optimized mode engineering, state-of-the-art hybrid modulators are showing comparable performance to monolithic ones, which may already be enough for most practical applications^{4,73,79,84–86}.

In addition, there are no TFLN-based lasers and photodetectors, but impressive progress has been made to integrate these missing functionalities on the TFLN platform. Optically pumped lasers have been achieved using rare-earth-ion-doped disk, ring and coupled-ring cavities^{171,172}. These implementations, however, suffer from low conversion efficiency (less than 1%) and limited power (of the order of microwatts), require an external pump laser (typically at 980 nm for Er^{3+} doping), and would eventually need selective doping to avoid ion-absorption-induced loss on other part of the photonic circuits.

Alternatively, hybrid or heterogeneous integration of high-performance III–V laser, gain or photodetector devices may provide a better solution for applications such as microwave photonics and telecommunications. The hybrid approach refers to placing a III–V gain chip next to the TFLN circuit. This configuration has proved particularly advantageous for injection locking of distributed feedback (DFB) lasers or realizing tunable and narrow-linewidth external-cavity lasers. Some notable examples of this approach include a narrow-linewidth O-band LN/III–V transmitter with wide wavelength-tuning range¹⁷³, coherent frequency-modulated continuous-wave lidar demonstration with an injection-locked DFB laser on a hybrid TFLN–SiN platform⁸, and a rapidly tuning integrated Pockel's laser¹⁷⁴. The laser performance is highly sensitive to the position and facet preparation of the two chips. Most experiments have the LN and III–V chips on separate mounts or even different stages, limiting the maximum optical power in the waveguides. A fully integrated flip-chip DFB laser directly bonded on TFLN addressed this issue and coupled more than 60 mW in the waveguide¹⁷⁵, followed by an improvement to 115 mW of optical power later¹⁷⁶.

The highest degree of integration can be achieved using heterogeneous techniques, including micro-transfer printing^{177,178} or direct bonding^{179,180}. In the heterogeneous scheme, the coupling between the III–V material and LN circuit can be potentially near unity using adiabatic tapers. However, at the moment all demonstrations have around 1 mW of optical power, mainly because of process imperfections and systematic misalignment. Heterogeneously integrated lasers typically use a polymer layer between the two platforms as a planarization and adhesion agent that compromises their thermal stability. Additionally, heterogeneous integration necessitates the simultaneous processing of multiple materials, leading to a more complex and constrained fabrication process. Recently, a proof-of-principle technique

realized scalable integration while circumventing the thermal issues of heterogeneous approach. Despite a complex fabrication process, the devices still suffer from relatively low optical output of around 2 mW (ref. 181). Heterogeneous integration of a photodetector on TFLN was demonstrated using adhesive bonding^{179,182,183}, featuring an 80 GHz bandwidth and 0.6 A W⁻¹ responsivity¹⁸² as well as 110 GHz bandwidth and 0.4 A W⁻¹ responsivity¹⁸³. More efforts and progress were made in the heterogeneous integration of photodetectors on TFLN using adhesive bonding^{179,183}. A comprehensive review of recent advancements on lasers and amplifiers with and on TFLN can be found elsewhere¹⁸⁴.

Emerging applications

With the recent rapid advancements in EO devices on TFLN, the field is transitioning from focusing on individual devices to comprehensive system-on-a-chip approaches. Although there are many expected applications of this platform, including those in telecommunications, data communications and microwave photonics, here we focus on several emerging applications and future opportunities.

Photonic computing and accelerators

Tightly integrated and power-efficient electronic and photonic systems can enable important advances and push the state of the art in optical communication systems. Furthermore, we envisage that such an integrated electronic–photonic system using TFLN will enable breakthroughs in optical computing and microwave photonics. With rapid advancements in artificial intelligence and development of deep neural networks, central processing units and graphics processing units alike are challenged by impending bottlenecks in memory and speed while maintaining small form factor and low power consumption. Recently, the photonics community has responded to these challenges with new architectures, specialized for accelerating the training and inference stages of such networks. For example, a pioneering work demonstrated audio recognition using a silicon-photonic chip¹⁸⁵. Following this demonstration, many photonic neural network accelerators emerged, including processors based on frequency combs¹⁸⁶, phase-change materials¹⁸⁷, current-controllable optical attenuators¹⁸⁸, Mach–Zehnder interferometer meshes¹⁸⁹, vertical-cavity surface-emitting laser arrays¹⁹⁰, optical diffraction¹⁹¹, and combinations with communications¹⁹², further pushing the boundaries of computational accuracy, speed and power consumption.

Recognizing the impact, the TFLN platform promises to provide orders-of-magnitude advantages in aspects of speed and power consumption and might play a key role in finally bringing photonic neural networks to market. The heterogeneous integration of lasers^{8,171–180} on TFLN provides high-power data channels, whereas EO^{2,7,150} and Kerr^{193–195} frequency combs provide potential for massively parallel frequency-division multiplexing across hundreds of channels. In each channel, high-speed EO modulators operating at CMOS-compatible voltages^{1,4,5} provide unprecedented levels of computational power and reconfigurability owing to the analog nature of optical processing, in addition to efficiently encoding and manipulating data with ultralow power consumption. Finally, the native $\chi^{(2)}$ nonlinearity of TFLN coupled with high-speed integrated III–V photodiodes^{179,182,183} presents an opportunity for on-chip nonlinear activation and opto-electronic conversion and detection, closing the loop for all-optical computation. As TFLN continues to mature in the coming years, we may witness a fully monolithic photonic neural network that is competitive with electronic approaches or even outclasses them.

Quantum photonics

The TFLN platform will play an important role in quantum photonics. The high carrier frequency of light allows both high-bandwidth quantum communication and the measurement of quantum states of individual photons^{196–199} because the energy corresponding to the carrier frequency is far above the thermal background. The properties of TFLN, including low loss, possibility of large-scale integration, and possibility for photon–photon interactions through EO and/or optical $\chi^{(2)}$, render it an attractive platform for generating and manipulating quantum states of light²⁰⁰. Although a single-photon nonlinearity is not currently efficient in TFLN²⁰¹, the possibility of beam splitting and high-bandwidth EO phase shifting makes it attractive for linear-optical quantum computing¹⁶⁴. Moreover, combs and modulators allow for computing protocols in the frequency domain^{57,159–163,202–208}, taking advantage of the multiplexing ability of light. Indeed, the coupled-ring system³ offers frequency-domain beam splitting without expanding the Hilbert space compared to using a single phase modulator²⁰⁹. High-bandwidth switching offered by TFLN is also a key ingredient for rendering inefficient spatially multiplexed heralded states quasi-deterministic and can be applied to photon sources and quantum communications^{210–212}.

The all-optical $\chi^{(2)}$ interactions on TFLN, often performed using periodic poling of TFLN²¹³, further enhances the EO platform. For example, it can be used for photon pair generation, squeezed light generation, and deterministic wavelength conversion of individual photons. These are of interest for photonic quantum computing, realization of quantum interconnects between devices that operate at different energy scales, upconversion detection^{214,215}, quantum sensing and computing using highly squeezed states^{216–218}, as well as realizing efficient frequency combs via cascaded $\chi^{(2)}$ interactions²¹⁹. Beyond EO and $\chi^{(2)}$, the piezoelectric coupling of TFLN has been harnessed for quantum interconnects between microwave photons and acoustic phonons, with applications in, for example, transduction or acoustic modulators²²⁰. Furthermore, one can embed rare earth ions into TFLN²²¹, adding the possibility of quantum emitters, storage, signal manipulation²²² and sensors to the range of functionalities offered by TFLN itself. Overall, the impact that TFLN has on quantum photonics with electro-optics is tremendous, and we have only recently begun to explore its potential.

Active topological photonics and non-Hermitian physics

Microwave-coupled optical lattices in frequency domain realized on TFLN^{3,7,10,17,22,23} are emerging as a promising platform to study topological photonics, non-Hermitian physics and microcavity sensing, for example. Combining TFLN with microcavity sensing^{223,224} is particularly promising for sensors that are both fast and highly sensitive. With the inherent strong piezoelectric interaction on TFLN, EO modulation could add a new control of degree of freedom to conventional acoustic sensors^{223,224}. Moreover, EO modulation on TFLN may enable the practical implementations of modulation-induced gauge fields for photons¹⁴⁴. Collectively, these advancements underscore the unique value of TFLN in offering additional degrees of freedom for ultrafast tuning for various optical systems, making it an ideal platform to study optical physics using waveguides and cavities and opening new directions for the manipulation and study of light.

Nonlinear photonics

The excellent $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities of TFLN allow the combination of parametric frequency-conversion processes with electro-optics,

opening multiple opportunities for exploring fundamental physics, developing novel devices and addressing practical applications. From a physics perspective, the combined $\chi^{(3)}$ -EO⁷ and $\chi^{(2)}$ -EO^{19,225} introduces extra degrees of freedom and interactions for synthetic EO lattice. For example, the coexistence of the EO and Kerr effects^{7,226} indicates that it is feasible to introduce the on-site nonlinear interactions via the Kerr effect in the synthetic EO lattice. This allows for the simulation of strongly correlated condensed matter models beyond the single electron approximation. From an applications standpoint, integrated nonlinear frequency combs are particularly noteworthy, covering applications such as tunable optical parametric oscillators^{225,227,228}, ultrastable microwave^{229,230} and millimetre-wave generation²³¹, optical frequency referencing^{232,233} and synthesis²³⁴, astrocombs^{235,236}, and ranging^{30,237}.

The potential of integrated nonlinear frequency combs is set to expand by leveraging the multiple strong nonlinearities on TFLN. For example, at the device level, TFLN facilitates novel comb generation that may harness the best properties of the EO and Kerr effects, such as EO-Kerr combs in the same resonator⁷, EO-tunable Kerr combs²³⁸, and cascading EO modulators with Kerr resonators²³⁹. At the system level, the combination of comb sources, modulators and periodically poled LN waveguides can benefit these applications. Furthermore, the integration of new gain technologies such as the reflective semiconductor optical amplifier¹⁷³ and ion-embedding methods such as erbium-doping of TFLN^{240–242} and thulium-doped cladding oxides²⁴³ for TFLN is promising. It may open exciting avenues in integrated lasers^{8,171–180} at exotic wavelengths and ultrafast, high-peak-power pulse generators⁹ and amplifiers^{243,244}. Such advances promise the deployment of chips for atomic physics at visible and near-ultraviolet²⁴⁵, fully integrated supercontinuum generation²⁴⁶ and biomedical imaging²⁴⁷.

Ultrafast control

TFLN offers advanced capabilities for ultrafast light control such as frequency tuning^{8,32,33}, beam steering and lidar systems^{30,31,248,249}, owing to its high EO coefficients, wide bandwidth and low optical loss. These properties enable ultrafast modulation and precise control over the spatial direction of light as well as the frequency of light, which are critical for applications such as advanced optical phased arrays. The integration on TFLN into compact devices allows for real-time, high-resolution beam steering without mechanical components, considerably enhancing the performance of lidar systems used in autonomous vehicles and environmental monitoring. Moreover, the tight optical confinement achievable in TFLN waveguides aids efficient phase shifting and amplitude control, enabling faster and more energy-efficient LiDAR signal processing. These advantages position TFLN as a key material for ultrafast control towards next-generation lidar technologies, offering improvements in accuracy, speed and scalability for a range of applications, including robotics and remote sensing.

In this Review article, we have discussed the basic principles and formalisms of electro-optics on TFLN and introduced key research topics in this field. Electro-optics on TFLN has developed a lot over the past five years. It is becoming a promising solution for the next-generation of EO interfaces, such as optical communications, microwave photonics and photonic computing, where efficient and high-bandwidth EO conversion are paramount. TFLN has the potential to provide novel EO devices and circuits as well as hybrid circuits that interface with all-optical nonlinearities and acousto-optic interactions. As it matures, we expect TFLN to become one of the most important platforms for integrated photonics.

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Author contributions

Y.H., D.Z., S.L., X.Z., Y.S., D.R., D.A., N.S. and M.L. researched data for the article. Y.H., D.Z., S.L., X.Z., Y.S., D.R., D.A., N.S. and M.L. contributed substantially to discussion of the content. Y.H., D.Z., S.L., X.Z., Y.S., D.R., D.A., N.S. and M.L. wrote the article with contributions from all authors. All authors reviewed and edited the manuscript before submission.

Competing interests

M.L. is involved in developing lithium niobate technologies at HyperLight Corporation. The other authors declare no competing interests.

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